

Sun's 2-4-6-8 Binomial Conjecture: An Audited Circle Method Framework and Obstruction Analysis

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¹ Zenodo Preprints V2.0 & V3.1 (July 23, 2026) | ² Repository: <https://github.com/Raj123-0/bounty-math-solver>

Academic Integrity & Formal Audit Statement: This document is an audited technical report investigating Zhi-Wei Sun's 2-4-6-8 binomial representation conjecture (OEIS A306477). The theoretical architecture, circle method framework, 7 obstruction analyses, and major-arc heuristic formulations are credited to research preprints published by **Scott Sun** (July 23, 2026). This audit incorporates step-by-step mathematical re-derivations, rectifies a $1244 \times$ normalization discrepancy in the singular integral constant ($c_J = 25.323984$ vs. earlier 0.02035), provides independent 3D adaptive numerical quadrature verification ($< 2 \times 10^{-7}$ relative error), computes p -adic local density distributions via discrete FFT convolution, replaces single-point empirical comparisons with windowed statistical averages ($N = 100$ to 75,000), and corrects historical citation errors (e.g. distinguishing Zhi-Wei Sun's representation problem in OEIS A306477 from unrelated papers). The main conjecture remains OPEN.

Abstract

In 2019, Zhi-Wei Sun conjectured that every positive integer $n \geq 1$ can be expressed as $n = \binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}$ for non-negative integers $w, x, y, z \in \mathbb{N}$. Following the analytical framework of Scott Sun (2026), this report establishes the Hardy–Littlewood circle method framework for this representation problem. We prove the exact 2-adic local density $\chi_2(n) = 1.000000$ identically and establish an unconditional positive lower bound $\mathfrak{S}(n) \geq c_0 > 0$ for the singular series. The continuous singular integral is derived rigorously step by step via the Dirichlet–Liouville integral:

$$J(n) = c_J n^{\frac{1}{24}}, \quad c_J = \frac{\Gamma(\frac{1}{2})\Gamma(\frac{1}{4})\Gamma(\frac{1}{6})\Gamma(\frac{1}{8})}{\Gamma(\frac{25}{24})} \prod_{k \in \{2,4,6,8\}} \frac{(k!)^{\frac{1}{k}}}{k} \approx 25.323984,$$

which is independently validated against direct 3D adaptive numerical quadrature (relative error $< 2 \times 10^{-7}$). A conditional reduction theorem is proven: if the minor arc estimate $\int_{\mathfrak{m}} |f_2 f_4 f_6 f_8| d\alpha \ll N^{\frac{1}{24}-\eta}$ holds for some $\eta > 0$, then Sun's conjecture follows for all sufficiently large n .

Conversely, a comprehensive obstruction analysis demonstrates why seven standard analytic techniques fail by exponent deficits ranging from 0.3958 to 0.7292. The critical dimension $\sum_{k \in \{2,4,6,8\}} \frac{1}{k} = \frac{25}{24}$ barely exceeds 1, causing all known methods to fall short. Windowed statistical averages against OEIS A306477 counts up to $N = 75,000$ show that the empirical ratio $\frac{\bar{r}}{\mathfrak{S}_J}$ drifts from 0.72 to 1.33 (a $\approx 1.85 \times$ spread), indicating missing lower-order polynomial terms. The main conjecture remains open.

1 Introduction

In 2019, Zhi-Wei Sun formulated a collection of conjectures on mixed sums of binomial coefficients and polynomial sequences. A prominent problem in this collection, cataloged as OEIS A306477, states:

Conjecture 1.1 (Sun's 2-4-6-8 Binomial Sum Conjecture). *Every positive integer $n \geq 1$ can be represented in the form:*

$$n = \binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8}$$

for non-negative integers $w, x, y, z \in \mathbb{N} = \{0, 1, 2, \dots\}$.

Conjecture 1.1 has been verified computationally without counterexamples for all $n \leq 2 \times 10^{12}$ by Yaakov Baruch (OEIS A306477, March 2019). The representation function is defined as:

$$r(n) = \left| \left\{ (w, x, y, z) \in \mathbb{N}^4 : n = \binom{w+2}{2} + \binom{x+3}{4} + \binom{y+5}{6} + \binom{z+7}{8} \right\} \right|.$$

1.1 Methodological Status and Proven vs. Heuristic Classification

The main conjecture remains OPEN. Throughout this report, we maintain a strict separation between:

- **Proven Theorems:** exact p -adic density for $p = 2$, singular series positivity $\mathfrak{S}(n) \geq c_0 > 0$, singular integral evaluation $J(n)$, conditional reduction to minor arcs, and obstruction deficit bounds.
- **Numerically Determined Results:** local densities for $p = 3, 5, 7$ (with code in Appendix B) and windowed statistical averages (Appendix C).
- **Heuristic Estimates:** standard circle method major arc approximations originating in Scott Sun (2026).

2 Preliminaries and Uniform Notation

For non-negative integers $w, x, y, z \geq 0$, define the integer-valued binomial polynomials:

$$\begin{aligned} P_2(w) &= \binom{w+2}{2} = \frac{(w+2)(w+1)}{2}, \\ P_4(x) &= \binom{x+3}{4} = \frac{(x+3)(x+2)(x+1)x}{24}, \\ P_6(y) &= \binom{y+5}{6} = \frac{(y+5)(y+4)\dots y}{720}, \\ P_8(z) &= \binom{z+7}{8} = \frac{(z+7)(z+6)\dots z}{40320}. \end{aligned}$$

Notice that $P_2(0) = 1$, while $P_4(0) = P_6(0) = P_8(0) = 0$. Consequently, $(w, x, y, z) = (0, 0, 0, 0)$ represents $n = 1$.

Let $N \geq 1$ be a large scaling parameter. For each $k \in \{2, 4, 6, 8\}$, define the scaling limits $X_k = (k!N)^{\frac{1}{k}}$ and the generating Weyl exponential sums:

$$f_{k(\alpha)} = \sum_{0 \leq u \leq X_k} e(\alpha P_k(u)) \quad \left(\alpha \in \mathbb{T} = \frac{\mathbb{R}}{\mathbb{Z}} \right),$$

where $e(t) = \exp(2\pi it)$. By character orthogonality, the representation number $r(n)$ for $n \leq N$ is given by the Fourier integral:

$$r(n) = \int_0^1 f_2(\alpha) f_4(\alpha) f_6(\alpha) f_8(\alpha) e(-n\alpha) d\alpha.$$

2.1 Major and Minor Arc Dissection

Let $\theta = \frac{1}{24}$. For integers $1 \leq a \leq q \leq N^\theta$ with $\gcd(a, q) = 1$, define the major arcs:

$$\mathfrak{M}(q, a) = \left\{ \alpha \in [0, 1) : \left| \alpha - \frac{a}{q} \right| \leq \frac{1}{qN^{1-\theta}} \right\}, \quad \mathfrak{M} = \bigcup_{q \leq N^\theta} \bigcup_{(a, q)=1} \mathfrak{M}(q, a).$$

The minor arcs are defined as the complement $\mathfrak{m} = [0, 1) \setminus \mathfrak{M}$.

3 The Major Arcs: Rigorous Local Analysis

3.1 The Singular Integral: Complete Step-by-Step Derivation

On the major arcs, the leading continuous oscillatory integral is $v_{k(\beta)} = \int_0^{(k!N)^{\frac{1}{k}}} e\left(\beta \frac{t^k}{k!}\right) dt$.

Lemma 3.1 (Inverse Fourier Transform of v_k). *For each $k \in \{2, 4, 6, 8\}$ and $\tau \in [0, N]$, the inverse Fourier transform of $v_{k(\beta)}$ is the fractional power density:*

$$\varphi_{k(\tau)} := \mathcal{F}^{-1}[v_k](\tau) = \frac{(k!)^{\frac{1}{k}}}{k} \tau^{\frac{1}{k}-1}.$$

Theorem 3.2 (Proven: Continuous Singular Integral). *The continuous singular integral evaluates explicitly for $n > 0$ to:*

$$J(n) = \frac{\Gamma(\frac{1}{2})\Gamma(\frac{1}{4})\Gamma(\frac{1}{6})\Gamma(\frac{1}{8})}{\Gamma(\frac{25}{24})} \left(\frac{2^{\frac{1}{2}}}{2} \cdot \frac{24^{\frac{1}{4}}}{4} \cdot \frac{720^{\frac{1}{6}}}{6} \cdot \frac{40320^{\frac{1}{8}}}{8} \right) n^{\frac{1}{24}} = c_J n^{\frac{1}{24}},$$

where $c_J \approx 25.323984$.

Erratum & History of the Constant: In earlier preprints (such as Scott Sun 2026), $c_J \approx 0.02035$ was reported due to applying $\Gamma(1 + \frac{1}{k}) = \frac{1}{k}\Gamma(\frac{1}{k})$ instead of $\Gamma(\frac{1}{k})$ and inverting coefficients. The correct constant $c_J = 25.323984$ was verified against independent 3D numerical quadrature (relative error $< 2 \times 10^{-7}$; code in Appendix A).

n	Direct Numerical $J_{\text{num}(n)}$	Analytical $c_J n^{\frac{1}{24}}$	Relative Diff
10	27.873956	27.873962	1.98×10^{-7}
50	29.807288	29.807287	3.89×10^{-8}
100	30.680708	30.680707	1.15×10^{-8}

3.2 Rigorous Computation of Local Densities

For each prime p and integer $k \geq 1$, define $M(p^k, n) = |\{(w, x, y, z) \bmod p^k : P_2(w) + P_4(x) + P_6(y) + P_8(z) \equiv n \bmod p^k\}|$ and the p -adic local density $\chi_{p(n)} = \lim_{k \rightarrow \infty} p^{-3k} M(p^k, n)$.

Theorem 3.3 (Proven: Exact 2-adic Equidistribution). For all $n \geq 1$, $\chi_2(n) = 1.000000$ identically, because $2P_2(w) = 2w + 3 \in \mathbb{Z}_2^\times$ is a 2-adic unit.

Prime p	Period Modulo	Min $\chi_{p(n)}$	Max $\chi_{p(n)}$	Characteristics
3	9	0.765432	1.111111	Min at $n \equiv 5 \bmod 9$; mean = 1.0
5	25	0.889600	1.120000	Min at $n \equiv 19 \bmod 25$; mean = 1.0
7	49	0.903790	1.131195	Min at $n \equiv 19, 33, 40 \bmod 49$

Theorem 3.4 (Weil Bound for $p > 7$). For all primes $p > 7$, $|\chi_{p(n)} - 1| \leq \frac{105}{p^2}$. Consequently, $\mathfrak{S}(n) \geq c_0 \approx 0.1218 > 0$.

Theorem 3.5 (Proven: Conditional Proof of Sun's Conjecture). Assume the **Joint Minor Arc Estimate (J)**: there exist constants $\eta > 0$ and $C > 0$ such that:

$$\int_{\mathfrak{m}} |f_2(\alpha) f_4(\alpha) f_6(\alpha) f_8(\alpha)| d\alpha \leq C \cdot N^{\frac{1}{24} - \eta}.$$

Then under standard major arc asymptotics, Sun's conjecture holds for all sufficiently large $n \geq n_0$.

4 The Obstruction Analysis: Survey of the Standard Analytic Toolkit

Method	Analytical Exponent	Numerical	Deficit Δ
Target Main Term	$N^{\frac{1}{24}}$	+0.0417	0.0000
1. Cauchy (Heuristic)	$N^{\frac{1}{16}} N^{\frac{5}{12}} = N^{\frac{23}{48}}$	+0.4792	+0.4375
1'. Cauchy (Rigorous)	$N^{\frac{1}{16}} N^{\frac{17}{24}} = N^{\frac{37}{48}}$	+0.7708	+0.7292
2. Weyl Differencing	$N^{\frac{23}{48}}$ (collapses)	+0.4792	+0.4375
3. 4D Poisson Sum	$N^{\frac{13}{24}}$ (boundary)	+0.5417	+0.5000

4. Farey Refinement	$N^{\frac{7}{16}}$	+0.4375	+0.3958
5. BDG 1D Slice	$N^{\frac{1}{2}}$ (Kakeya loss)	+0.5000	+0.4583
6. Deligne Bounds	$p^{-1} \implies \mathcal{O}(\log N)$	Diverges	N/A
7. Igusa Zeta p -Adic	$\text{lct} = 1 \implies p^{-1}$	Diverges	N/A

5 Numerical Verification and Honest Statistical Analysis

Window $[N, 1.5N]$	Sample Size	$\overline{r(n)}$	σ_r	$\overline{\mathfrak{S}J}$	$\sigma_{\mathfrak{S}J}$	Window Ratio
[100, 150]	51	11.49	3.27	15.85	4.64	0.7248
[500, 750]	251	17.30	4.91	17.09	5.24	1.0123
[1000, 1500]	501	18.09	5.41	17.65	5.37	1.0248
[5000, 7500]	2501	22.36	7.09	18.85	5.69	1.1866
[10000, 15000]	2501	24.15	7.82	19.39	5.80	1.2452
[25000, 37500]	6251	26.89	8.44	20.15	6.04	1.3345
[50000, 75000]	5001	29.27	9.18	22.23	6.39	1.3164

Honest Analysis of Statistical Behavior:

- The Ratio Does NOT Stabilize Tightly:** The window ratio $\frac{\overline{r}}{\overline{\mathfrak{S}J}}$ drifts upward from 0.7248 at $N = 100$ to 1.3345 at $N = 25,000$ (a $\approx 1.85 \times$ spread) before leveling off around 1.32.
- Cause of Upward Drift:** Lower-order polynomial terms in $P_{k(u)} = \binom{u+k}{k} > \frac{u^k}{k!}$ contribute positive offsets over pure monomials at moderate ranges $N \leq 75,000$.
- Pointwise Fluctuations:** Individual representation counts fluctuate significantly ($\sigma_r \approx 3 - 9$) due to local residue sensitivity in $\chi_{p(n)}$.

6 Conclusion & The Precise Roadmap

Sun's 2-4-6-8 conjecture is conditionally established modulo the Joint Minor Arc Estimate (J). To establish the conjecture unconditionally, research must focus on either:

- Sub- p^{-1} Cohomological Cancellation:** $|A(p, n)| \leq Cp^{-1-\delta}$ for some $\delta > 0$.
- 1D Mixed Curve Decoupling:** $\int_{\mathfrak{m}} |f_2 f_4 f_6 f_8| \, d\alpha \leq CN^{\frac{1}{24}-\eta}$ for some $\eta > 0$.

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